

Dr. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Semester Examination- 2014-15

Class: B.Tech.E&TC

Semester:- VIII

Subject: Embedded System Design (old)

Time:3 Hrs

Date: 16/5/15

Max. Marks: 50

INSTRUCTIONS TO THE STUDENTS:-

1. Attempt at least ONE question from each GROUP & attempt anyone from remaining.
2. Assume the suitable data wherever necessary and mention it clearly.
3. Draw necessary diagram wherever required.

GROUP-1

- Q.1 A) Write a note on general purpose processor in embedded system. (05)
- B) Distinguish between ARM instruction set and THUMB instruction set. (05)
- Q.2 A) Write a note on basic interrupt stack design and implementation. (05)
- B) Write a note on instruction pipeline in different ARM core and explain any one with example. (05)

GROUP-2

- Q.3 A) What is embedded system? List out different design metrics in embedded system. (05)
- B) How the Fast Context Switch Extension uses page tables and domains in MMU. (05)
- Q.4 A) State the rules for the RISC Design Philosophy. (05)
- B) Explain software interrupt instruction with example. (05)

GROUP-3

- Q.5 A) Explain Protected Regions in memory protection unit. (05)
- B) Explain the procedure of defining Region size and location in memory protection unit. (05)
- Q.6 A) Explain mapping of a task in virtual memory to physical memory using a relocation register. (05)
- B) Write a note on translation look-aside buffer. (05)

GROUP-4

- Q.7 A) Discuss the goals of OS. (05)
- B) Define the following Terms. (05)
- I) Process II) Task III) Threads
- Q.8 Explain embedded hardware component for smart card system. (10)